Australia GDP Forecasting

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Research Question and Data

- **Objective**
  - Build and compare models for quarterly Australian year-on-year real GDP growth, using recursive one-step-ahead forecasts over 2016Q1–2025Q4.
- **Sample**
  - 1993Q1–2025Q4
- **Sources**
  - Australian Bureau of Statistics and Reserve Bank of Australia.
- **Target**
  - Real GDP growth, year-on-year log change, percent.
- **Main predictors**
  - Inflation, cash rate, change in cash rate, unemployment.

Descriptive Analysis

- Pre-COVID GDP growth is comparatively stable, with moderate cyclical movement around the GFC.
  - The 2020 contraction and 2021 rebound are the dominant outliers.
- Inflation and the cash rate shift sharply after 2021.
- Unemployment declines until the early 2000s, then fluctuates around a lower level, with a temporary COVID spike.
- ADF tests reject a unit root for GDP growth, inflation, unemployment, and the cash-rate change.
  - Thus, we use these variables for our main specifications.

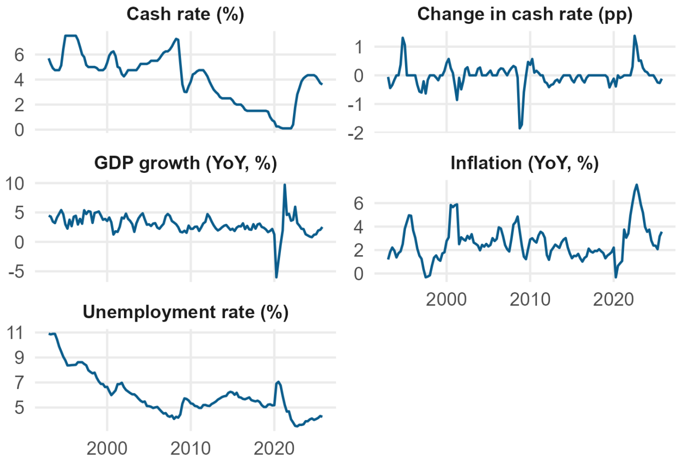

Figure 1: Australian macroeconomic series

GDP Growth Persistence

- GDP growth has strong first-lag persistence.
- The ACF decays gradually, supporting autoregressive dynamics.
- PACF spikes also appear at seasonal lags.
- Since the target is year-on-year growth, lags 4 and 8 are especially important due to base effects.
  - So, we use model specifications that allow seasonal residual dynamics.

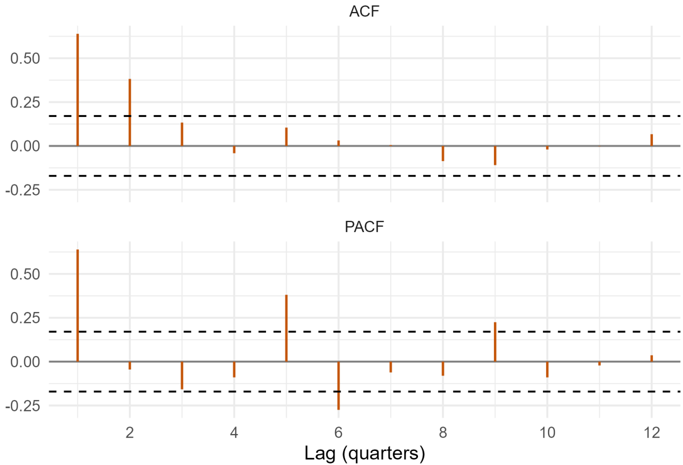

Figure 2: GDP growth ACF and PACF

Breaks, Outliers, and Controls

- For stability checks, we use general AR(2), AR(4), ARDL(2,2), and ARDL(4,4) models estimated on 1993Q1–2015Q4.
- Chow tests around 2008Q3–2009Q1 indicate GFC-related instability, especially in richer ARDLs.
    - However, variance tests suggest that part of the finding is driven by crisis period volatility, not only coefficient instability.
- Non-benchmark specifications include a post-2008Q4 intercept dummy to account for potential breaks.
    - Full interactions were excluded for parsimony.
- Post-dummy CUSUM paths remain inside 5% bands; Bai-Perron tests do not show strong evidence for another break.
- For the full sample, main residual outliers are located in the COVID period.

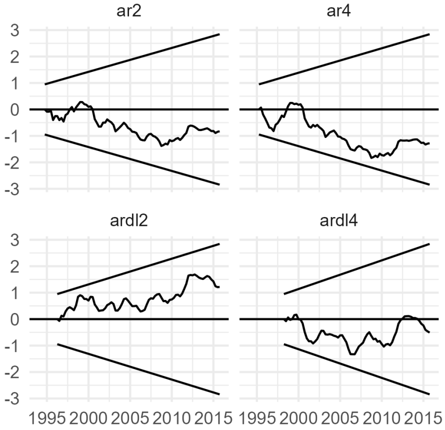

Figure 3: Recursive CUSUM after GFC dummy

Candidate Model Families

- **AR benchmark:** AR(2).
- **VAR benchmark:** VAR(2) with GDP growth, inflation, and the cash rate.
- **VAR robustness:** VAR(2) with the cash-rate change.
- **ARDL:** GDP growth on its own lags plus lagged macro predictors.
- **ARDL-ARMA:** dynamic regression with ARMA residuals, including lag-4 seasonal terms.
- **ARMAX:** ARMA model with exogenous macro predictors and seasonal errors.

- Why seasonal ARMA errors?
  - For year-on-year growth, a shock in quarter $t-4$ mechanically affects current growth.
  - For example, the COVID collapse in 2020Q2 creates a large 2021Q2 rebound effect.
  - Seasonal ARMA errors target this lag-4 dependence.

Model Selection Strategy

1. Estimate broad ARDL, ARDL-ARMA, and ARMAX model sets.
   - Permute over variables, lags (up to 4), ARMA components.
   - Sample: 1993Q1–2015Q4.
2. Keep top models by BIC and AIC.
3. Check residual diagnostics, flag violations.
4. Produce recursive one-step forecasts for 2016Q1–2025Q4.
5. Rank finalists by MSFE/MAFE and compare with AR(2), VAR(2).
   - Forecast accuracy is the final criterion because in-sample fit can overfit.

Check/Test	Role
BIC	Parsimonious shortlist
AIC	Richer dynamic shortlist
Ljung-Box/BG	Serial correlation
BP/ARCH	Residual heteroskedasticity
Jarque-Bera	Residual normality
Roots/eigenvalues	Dynamic assumptions

Residual Diagnostics for Finalists

Model	Variables	LB/BG(4) p	LB/BG(8) p	BP/ARCH p	JB p
AR2	GDP lags 1–2	0.001	0.001	0.028	0.360
VAR2	GDP, inflation, cash rate lags 1–2	0.009	0.009	0.010	0.000
ARDL5201	GDP(1), infl.(1)	0.000	0.002	0.462	0.728
ARDL_ARMA5250	GDP(1), infl.(1), Δrate(1), unemp.(1); SMA(4)	0.534	0.334	0.925	0.684
ARDL_ARMA5010	GDP(1), unemp.(1); SMA(4)	0.623	0.483	0.929	0.633
ARMAX1450	AR(1), infl.(1), Δrate(2), unemp.(1); SMA(4)	0.207	0.147	0.884	0.894

Note: Ljung-Box p-values are reported for univariate models; multivariate Breusch-Godfrey p-values are reported for VARs.

- AR(2), VAR(2), and plain ARDL models have residual problems, especially autocorrelation.
- ARDL-ARMA and ARMAX models remove the main seasonal residual autocorrelation.
    - Some seasonal MA roots are close to one: models may not have a structural interpretation, but can still be used for forecasting.

Economic Hypotheses and Block Tests

Model	GDP persistence	Inflation block	Monetary-policy block	Unemployment block
AR2	0.687 ($p < 0.001$)	–	–	–
VAR2	0.514 ($p < 0.001$)	−0.213 ($p < 0.001$)	0.162 ($p = 0.051$)	–
ARDL5201	0.427 ($p < 0.001$)	−0.235 ($p = 0.009$)	–	–
ARDL_ARMA5250	0.508 ($p < 0.001$)	−0.157 ($p = 0.007$)	−0.130 ($p = 0.281$)	0.082 ($p < 0.001$)
ARDL_ARMA5010	0.659 ($p < 0.001$)	–	–	0.091 ($p < 0.001$)
ARMAX1450	0.511 ($p < 0.001$)	−0.332 ($p < 0.001$)	0.148 ($p = 0.018$)	0.150 ($p < 0.001$)

Note: reported values are sums of lag coefficients; the p-values test whether each lag block is jointly zero. For ARMAX, GDP persistence refers to the AR component.

- The results should not be interpreted causally.
- GDP persistence is positive and strongly significant in all finalists.
- Inflation is usually negative and statistically relevant.
- Monetary-policy effects are less robust across specifications.
- The positive and significant unemployment coefficient likely captures cyclical mean reversion after periods of weak growth.

Recursive Forecast Design

- **Recursive setup**
  - Evaluation period: 2016Q1–2025Q4.
  - Models are re-estimated recursively using an expanding window before each forecast date.
  - E.g., the first forecast uses data for 1993Q1–2015Q4.
- **Evaluation tools**
  - MFE for average bias and statistical bias test.
  - MSFE, MAFE for forecast loss.
    - Separately look at the non-COVID period, excluding 2020Q1–2022Q4 from the loss calculation.
    - Recursive estimation still uses all historical data available at each forecast date.
  - Forecast-error serial correlation tests for weak efficiency.
  - DM and HAC loss-difference tests versus AR(2) and VAR(2).

Recursive One-Step-Ahead Forecasts

- All models miss part of the 2020 collapse and 2021 rebound.
- Benchmarks adjust slowly around COVID.
- Seasonal ARMA-error models track the rebound and later normalization more closely.

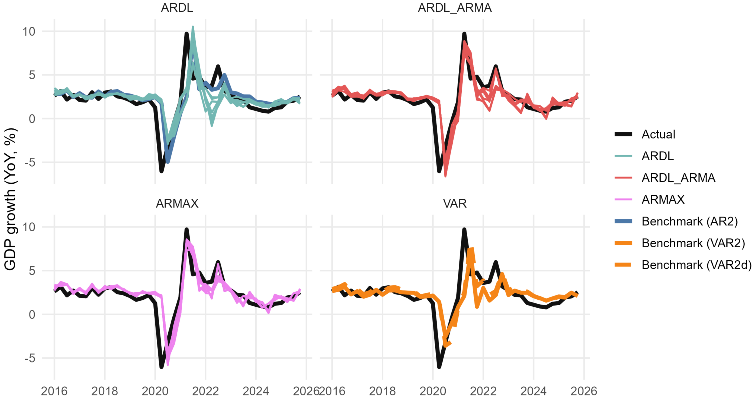

Figure 4: Recursive one-step-ahead forecasts by model family

Cumulative Squared Forecast Errors

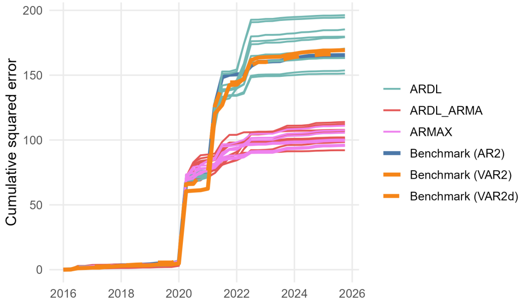

Figure 5: Full evaluation sample

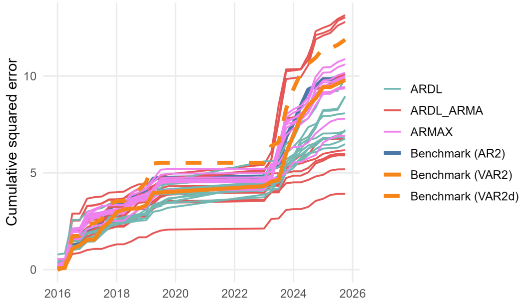

Figure 6: Excluding 2020Q1–2022Q4

- Full-sample losses are dominated by COVID-driven base effects.
- Excluding COVID, the ranking is less extreme, but ARDL-ARMA models still compare favorably.

Forecast Quality of Finalists

Model	MFE	MSFE	MAFE	Bias test p	LB(4) p	LB(8) p
AR2	-0.253	4.136	1.158	0.299	0.033	0.024
VAR2	0.007	4.234	1.170	0.984	0.208	0.046
ARDL5201	-0.033	3.782	1.006	0.918	0.225	0.220
ARDL_ARMA5250	-0.162	2.515	0.832	0.418	0.802	0.592
ARDL_ARMA5010	-0.138	2.302	0.755	0.456	0.771	0.507
ARMAX1450	-0.256	2.480	0.858	0.250	0.923	0.650

- On average, most models tend to overshoot GDP growth, but any bias is statistically insignificant.
- Some evidence that benchmark model errors may be inefficient; accounting for additional variables and seasonal characteristics improves efficiency.

Forecast Accuracy of Finalists

Model	MSFE	MSFE/AR2	MSFE/VAR2	MSFE/AR2 ex. COVID	DM vs AR2 p	HAC loss vs AR2 p	DM vs VAR2 p	HAC vs VAR2 p
AR2	4.136	1.000	0.977	1.000	–	–	0.436	0.417
VAR2	4.234	1.024	1.000	0.985	0.564	0.583	–	–
ARDL5201	3.782	0.914	0.893	0.652	0.249	0.163	0.132	0.189
ARDL_ARMA5250	2.515	0.608	0.594	0.522	0.118	0.124	0.133	0.128
ARDL_ARMA5010	2.302	0.557	0.544	0.394	0.089	0.098	0.104	0.100
ARMAX1450	2.480	0.599	0.586	0.696	0.106	0.112	0.121	0.111

Note: DM / HAC loss p-values are one-sided.

- **Best forecast model:** ARDL_ARMA5010; MSFE is 44.3% below AR(2).
  - ARDL_ARMA5010: GDP growth and unemployment lags with seasonal MA(4) error dynamics.
  - DM/HAC evidence is marginal, so the improvement is economically meaningful but statistically tentative.
    - DM/HAC p-values are anyway treated as indicative because some comparisons are close to nested and the evaluation sample is short.
- **Richer economic alternative:** ARDL_ARMA5250; includes GDP, inflation, monetary policy, unemployment, and seasonal MA errors.
- **Best ARMAX**: ARMAX1450: similar accuracy with ARMAX structure.

Conclusion, Caveats, and LLM Use

Main conclusion

- Models that allow seasonal ARMA residual dynamics outperform the AR(2) and VAR(2) benchmarks in MSFE terms.
- The differences are largest during COVID but remain visible in the non-COVID evaluation.

Economic interpretations

- GDP growth is persistent.
- Inflation and unemployment add predictive content.
- Cash-rate effects are less stable.
- Seasonal base effects are central for year-on-year GDP growth.

Caveats

- Only 40 forecast observations.
- COVID dominates full-sample loss.
- Some MA roots are near the unit boundary.
- Forecast gains are not strongly significant at conventional levels.

LLM usage disclosure: Codex was used for the initial code scaffold, but the code was later rewritten. ChatGPT was used for conceptual exploration, clearing up implementation questions, and sanity-checking the results. Final modelling choices, code execution, and reported results were checked by ourselves.